



RESEARCH

# Variation in coral reef fisheries production, employment, and living wage goals

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**Abstract** How fish production differs between reef environments, a fringing and island reef system, was explored for their influences of effort–stock–catch–revenue relationships and sustainability. Fisheries-independent fishable biomass, recovery rates, taxonomic composition, and fisheries-dependent metrics of effort, catch rates, and revenue were analyzed in the high production ocean exposed fringing reef and the low production ocean-sheltered island reef environments. Recovery rates of fish biomass in closures and catch rates supported the differential production predictions with the fringing reef fish ecological biomass-surplus production rate  $r$  of  $0.22 \pm 0.12$  ( $\pm 95\%CI$ ) and the island reefs of  $0.07 \pm 0.03$ , which potentially produces a fisheries capture of  $5.6 \pm 0.91$  and  $2.8 \pm 0.3$  tons/km<sup>2</sup>/y estimated from  $B_{msy}$ , respectively. Both reef types had high effort and yields below that estimated from  $B_{msy}$ , which indicated overfishing by both fisheries-independent and -dependent metrics. Multiple lines of evidence indicate that fisheries catch was bounded by minimum acceptable or opportunity cost income thresholds rather than the reef's natural productivity. For example, the higher production fringing reefs had lower biomass, exhibited higher production/biomass (P/B) turnover rates, and faster declining per area yields than the lower production island reefs. Fishing effort declined over

the ~34-month study period to stabilize catch per unit effort (CPUE) in both reef types to income values between Kenya's national poverty and minimum individual income thresholds. The more productive fringing reefs had a greater yield deficit than the sheltered reefs and would require a larger reduction in daily fishing effort (40% versus 15%) to achieve sustainability at censused stock levels. The less productive island reefs had considerable disproportionate losses of key vulnerable schooling taxa associated with a 25% yield deficit relative to the predicted potential yield. Protecting fish aggregations in the island and reducing fishing effort in both reef systems should increase the catch production and fisher's community wealth. Stock recovery required to recover ecological production and employment capacity is likely prevented by rising fish prices and resource extraction subsidies that have likely slowed the rate of decline in fishing effort.

**Dhahania** Kutofautiana kwa uzalishaji wa samaki katika miamba ya mikoani na visiواني kulichunguzwa kwa kuan-gazia mahusiano kati ya juhudi-uzani wa samaki-mapato ya kilo-mapato ya pesa na uvuvi endelevu. Uzani wa samaki wanaoweza kuvuliwa, viwango vya urejeshaji, jamii ya sam-aki, vipimo vya juhudi, viwango vya kuvua, na mapato ya pesa vilichambuliwa katika maeneo ya miamba ya mikoani, yenye uzalishaji mkubwa, na miamba ya visiواني iliyo na uzalishaji mdogo. Viwango vya urejeshaji wa uzani wa sam-aki katika hifadhi zisizovuliwa, na viwango vya kuvuliwa vilithibitisha utabiri wa utofauti katika uzalishaji wa  $0.22 \pm 0.12$  ( $\pm 95\%CI$ ) katika miamba ya mikoani na  $0.07 \pm 0.03$  visiواني, ambayo inaweza kutoa tani  $5.6 \pm 0.91$  na  $2.8 \pm 0.3$ /km<sup>2</sup>/y za uzani wa kiwango cha juu cha mavuno endel-evu mtawalia. Miamba yote miwili ilikuwa na juhudi kubwa na mavuno chini ya uzani wa kiwango cha juu cha mavuno

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endelevu, inayoonyesha uvuvi kupita kiasi. Ushahidi mwingi unaonyesha kuwa uvuvi umelenga kiwango cha chini cha pesa badala ya uzalishaji asilia miambani. Kwa mfano, miamba ya mikoani yenye uzalishaji mkubwa ilikuwa na uzani mdogo wa samaki, viwango vya juu vya ubadilisho (uzalishaji kugawa na uzani wa samaki), na wepesi wa upungufu wa kilo kwa eneo ikilinganishwa na miamba ya visi-wani yenye uzalishaji mdogo. Juhudi za uvuvi zilipungua katika kipindi cha miezi 34 ya utafiti ili kuimarisha mapato ya kilo ya samaki kwa mvuvi katika miamba yote miwili kati ya kigezo cha umaskini nchini na kiwango cha kujitegemea binafsi. Miamba ya mikoani yenye uzalishaji zaidi ilikuwa na upungufu mkubwa wa kilo kwa eneo kuliko miamba ya visi-wani. Miamba hii inahitaji upungufu mkubwa wa juhudi za uvuvi kwa siku (40% dhidi ya 15%) ili kufikia uendelevu katika uzani wa samaki uliobainishwa. Miamba ya visi-wani isiyokuwa na uzalishaji mkubwa ilionyesha kutokuwa na uwiano katika jamii ya samaki wanaojikusanya waliohatarini, jambo lililochangia upungufu wa 25% ya mapato ya kilo kwa eno ikilinganishwa na kilo kwa eneo yanayotarajiwa. Kulinda mkusanyiko wa samaki katika miamba ya visi-wani na kupunguza juhudi za uvuvi katika miamba yote miwili inapaswa kuongeza uzalishaji wa samaki na utajiri. Urejeshi wa samaki unaohitajika kurejesha uzalishaji na utajirii huenda unazuiliwa na kuongezeka kwa bei ya samaki na ruzuku zinazochangia uvuvi mwingi zinazopunguza hitaji la kupunguza juhudi za uvuvi.

**Keywords** Africa · Balanced harvest · Maximum sustained yield, poverty thresholds · Social–ecological interactions · Sustainability

## Introduction

Tropical fishers are experiencing wide-scale poverty that is not easily rectified by existing management (Giron-Nava et al. 2021; Teh et al. 2024). Coral reefs are productive and species diverse but still widely overfished and yet the recommended sustainable catch is poorly known (Nash and Graham 2016; McClanahan 2022). Moreover, incomplete and missing data and catch metrics create sources of variation that are impeding good decision making (Pauly et al. 2013; Pauly and Zeller 2014). Therefore, losses of food security, biodiversity, and ecological processes continue to decline in these tropical fisheries (Cinner et al. 2016; Barclay et al. 2023). Finally, fisheries goals that focus on providing welfare as opposed to wealth creation have consequences for governance systems and the recommended policies applied to these fisheries (Bene et al. 2010; Giron-Nava et al. 2021; Basurto et al. 2025). The consequences are a weak understanding of the production potential, value of coral reef fish fisheries, and particularly developing specific benchmarks

for yields, wealth, employment, and sustainability (Bavinck et al. 2024).

A focus on a limited number of production, catch metrics, and goals has impeded a rigorous understanding of sustainable coral reef fisheries. For example, total daily landed catch was one of the first and most frequently collected measures of fisheries production (Dalzell 1996; Samoilys et al. 2017). However, short-term total catch, by itself, cannot reliably evaluate and recommend sustainable catch benchmarks (Pauly et al. 2013). More stock and catch metrics and associated goals and tradeoffs must be considered when making policy and management decisions (Bavinck et al. 2024). For example, knowledge of fishing effort, fished area, and sampling effort are among the important metrics frequently missing among national and international fisheries statistics (Pauly and Zeller 2014, 2016). During periods of increasing catch, missing metrics combined with reliance on equilibrium catch assumption methods has prevented accurate estimates of yields and sustainability (McClanahan and Azali 2020). Many catch models overestimate yields because unfished stock biomass, variable production rates, fish pricing effects, time lags, stock and production deficits measures and their interactions are missing (Dao et al. 2023; Edgar et al. 2024). Further, combining a welfare policy approach, underestimating catch, and overestimating yield potential has promoted decisions to increase subsidies, employment, and fishing effort. The outcome has contributed to the widespread and persistent poverty in tropical fisheries nations (Giron-Nava et al. 2021; Teh et al. 2024). Therefore, resolving these many fundamental fisheries science issues of variability in production, effort, yield, income, and their tradeoffs is required for recommendations that can provide ecological and social sustainability goals.

The ability to estimate stock biomass and production has increased with underwater visual census methods. Consequently, there are now numerous estimates of coral reef fish biomass and some proxies for fisheries production (McClanahan et al. 2007a, b, 2019; MacNeil et al. 2015; Zamborain-Mason et al. 2023; Caldwell et al. 2024). These estimates have helped to envelope potential biomass and production estimates, but the variation and its causes are not well understood. For example, variance is often as high as mean yields, which challenges making specific local catch rate estimates and recommendations (McClanahan 2018a,b). Production rates ( $r$ ) are the single best predictor of yields and the largest source of variation but, empirical estimates of the multispecies  $r$  with confidence intervals are few, but they can range from ~0.05 to 0.30 (McClanahan 2018b; McClanahan and Azali 2020).

Variation in production is influenced by environmental factors, fish community composition, and associated species life histories (Morais and Bellwood 2018; McClanahan et al. 2025). For example, species and multispecies biomass

and production are influenced by fishing effort, water temperature, reef area, seawater productivity, and depth (Morais and Bellwood 2020; McClanahan et al. 2023). Models based on species life history have shown that species composition and fishing selectivity influence production (Morais and Bellwood 2019; Morais et al. 2023). For example, remote wilderness areas have large-bodied fish with high biomass, but slow growing fish with low long-term sustainable fisheries production potential compared to well managed fished seascapes with more mixed life histories (McClanahan et al. 2022, 2025). Therefore, integrating environmental, taxonomic, and fishing effort variability should improve production and sustainability variability estimates.

A key scientific question for the sustainable use of coral reef fisheries is how unfished biomass ( $B_0$  or  $K$ ) and production ( $r$ ) are influenced by fish life histories and environmental and geographic conditions. A related problem is how environmental variability will influence the accuracy of using multispecies or community models for estimating yields in species diverse coral reefs. While unfished biomass is the more easily estimated metric, data compilations have uncovered large differences between unfished marine reserves among fished seascapes and remote wilderness (McClanahan et al. 2022). For example, reef fish biomass in wilderness reefs is double measures in fully closed marine reserves. Unfished biomass estimates must therefore consider the seascape size and effect on the maintenance of fish life histories and biomass (Edgar et al. 2014). Multispecies biomass-production estimates suggest that  $r$  and  $B_0$  are modified by the size and intensity of fishing in the larger seascapes and are inversely correlated (McClanahan and Azali 2020). Therefore, variability in production should reflect interactions between the environment, taxa, and the intensity and selectivity of fishing effort.

The goal of the study was to contribute to a better understanding of the sources of variation in fisheries production, revenue, and employment capacity in two large coral reef seascapes of East Africa (Fig. 1). Specifically, large open-ocean exposed fringing and ocean-sheltered lagoonal island reef environments were compared using many of the same indices. We hypothesized that variations in geography and ocean exposure would influence biomass and production values, and that these would subsequently be reflected in differential fisheries effort-catch patterns between environments. Specifically, that the sheltered island environment would have lower production and effort than the ocean exposed fringing reef environment. The differences would be expected to lead to different proxies of sustainability and overfishing, such as fishing effort, yields relative to an ecologically determined maximum sustained yields (i.e.,  $B_{msy}$ ), and employment capacity and revenues. Secondly, environmental and gear use differences in the selective capture of taxonomic groups would potentially modify production

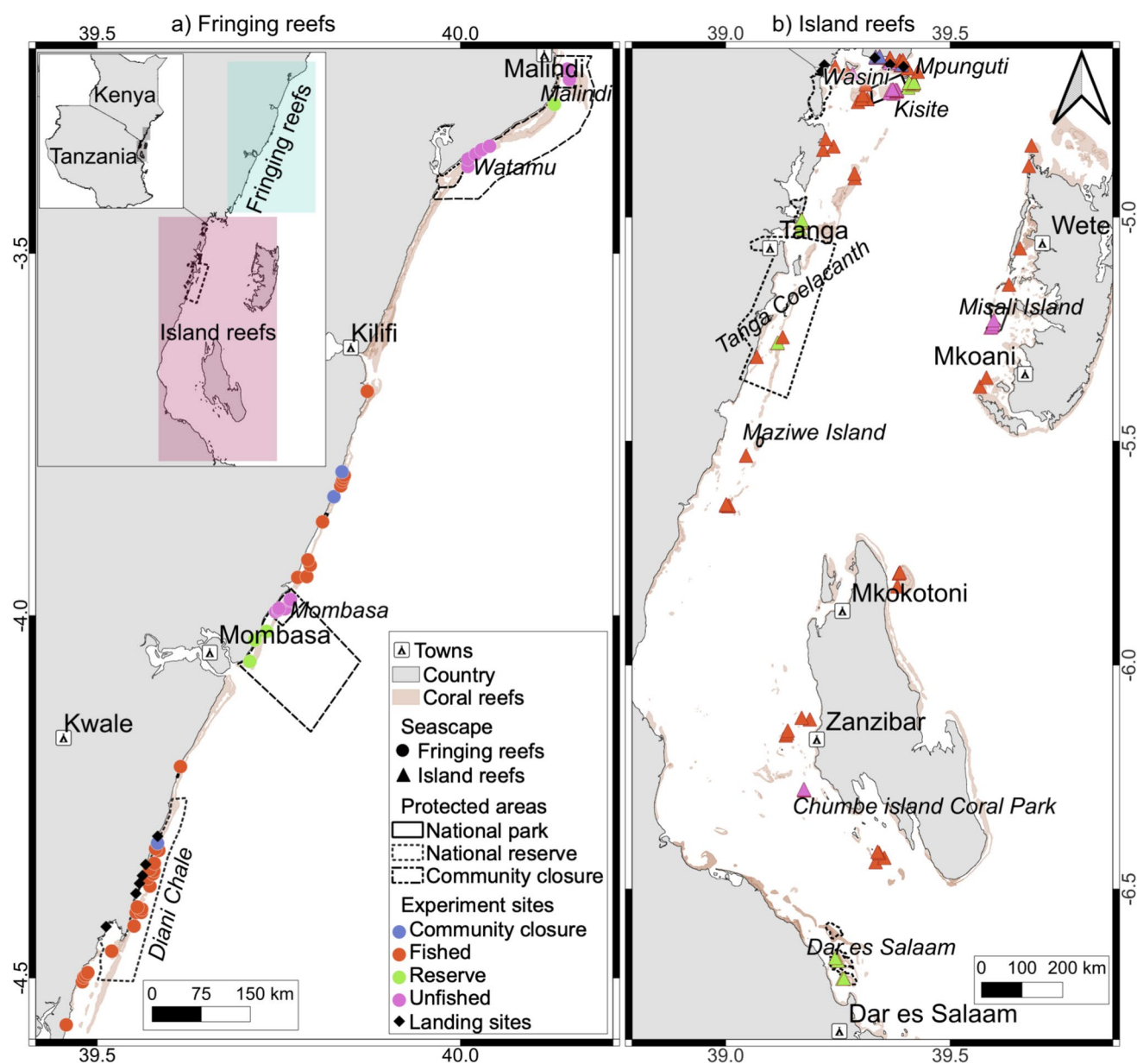
potential in both environments. Significant deviations from taxonomic proportionality would influence the community or multispecies production rates, the community  $B_{msy}$ , and fishing effort recommendations (Cunningham et al. 2009; McClanahan et al. 2025).

The implications of comparing these environments is that sustainability estimates would be different and potentially change with fisheries production potential. Increasing fishing effort and selectivity of the captured fish would cause production to deviate from either a uniform production (i.e., a single  $r$  value) or a “balanced harvest” scenario where taxa are captured in proportion to their production rates (Jacobsen et al. 2014). Therefore, key questions that will arise from comparing ecological production is if the community biomass surplus production models can reliably predict actual and MSY catches? Moreover, can biomass recovery and surplus production methods guide the goals of achieving proportional catch and maximized yield potential? Or, given that fishing can have strong selective forces on natural ecosystems, does ecological production fail to provide accurate and realistic management advice? Finally, should natural recovery after the cessation of fishing be a proxy for fisheries production and maximization estimates? The implications of this variability and uncertainty in yields are discussed in terms of the subsequent recommendations for sustainable fisheries management and poverty alleviation in coral reef fisheries.

## Materials and methods

### Study design overview

Reported measurements were intended to evaluate censused and ecologically sustainable yield estimates in large-scale fringing and island reefs geographies based on fisheries-independent biomass assessments and fisheries-dependent catch methods. Fisheries-independent methods included visual census of fish stocks by one observer (T.R. McClanahan) in a variety of management systems (McClanahan et al. 2007a, b). The time since closure to fishing in high compliance closure reefs was used as an estimate of biomass recovery ( $r$ ) or ecological production (Fig. 1; Table 1). Evaluations included recovery of biomass after closure to fishing (space-for-time evaluations) to estimate  $r$  and  $B_0$  based on multiple closures in each of the two geographic systems. Secondly, fisheries catch landed at six fringing and five island reefs were compared over  $34 \pm 1$  months to determine fisheries effort and production trends relative to expectations of the surplus ecological production from fishable biomass. The areas of the landing sites' fishing grounds were estimated via a collaborative community mapping process, which



**Fig. 1** Map of the East African coastline and coral reefs (in orange). Study sites showing **a** Kenya's southern fringing reefs and **b** the Kenya-Tanzania marine lagoon and island reefs. Shown are the studied fish landing sites and reef locations censused for fish biomass.

Official national and reserve park designations are shown as well as our management classification of sites as community closures, fished, reserves (restricted fishing gear), and unfished. These designations were used to estimate biomass (Table 1) and recovery rates (Fig. 2)

allowed estimates of per area efforts and yields (McClanahan and Kosgei 2023). Thirdly, the prices of fish and fishing revenue per fisher were calculated to evaluate the fisher's wages. Finally, comparisons of fish biomass in unfished and fished seascapes provided information on the potential for capture selectivity and the resistance of species to fishing.

## Field sampling

### Fish stock biomass

Fish stocks were sampled using visual underwater counts to estimate biomass over time in fisheries closures and to estimate fishable stock weights in four different fisheries

**Table 1** Summaries of biomass and yield estimates for fringing and island reefs for (a) censused biomass (mean kg/ha  $\pm$  95% CI, sample sizes) by management systems with Kruskal–Wallis test of significance between seascapes, (b) best fit logistic or surplus production model recovery information, (c) surplus yield estimates (tons/km<sup>2</sup>/y)

from censused biomass, (d) daily yield and revenue under censused and fish biomass ( $B_{msy}$ ), and (e) fishing effort estimates that achieve poverty and living wage thresholds at censused biomass modeled from daily per fisher catch and revenue as a function of fishing effort

| Fish census information                                                                 | Fringing reef                      | Island reef             | Kruskal–Wallis test                                                        |
|-----------------------------------------------------------------------------------------|------------------------------------|-------------------------|----------------------------------------------------------------------------|
| (a) Censused fishable biomass (kg/ha)                                                   | Mean $\pm$ 95% CI (sample size, n) |                         | ChiSq; Prob > ChiSq                                                        |
| National/private park closures                                                          | 972.3 $\pm$ 87.6 (90)              | 1057.9 $\pm$ 142.7 (38) | NS                                                                         |
| Marine reserves                                                                         | 131.8 $\pm$ 27.6 (28)              | 380.6 $\pm$ 67.6 (47)   | 32.8; < 0.0001                                                             |
| Community closures                                                                      | 265.2 $\pm$ 58.8 (23)              | 396.6 $\pm$ 66.6 (60)   | NS                                                                         |
| Fished reefs                                                                            | 132.3 $\pm$ 25.3 (122)             | 214.4 $\pm$ 77.1 (10)   | 66.6; < 0.0001                                                             |
| (b) Biomass recovery information                                                        |                                    |                         |                                                                            |
| Equilibrium biomass, $B_0$                                                              | 1012 $\pm$ 84.81 (85)              | 1606.0 $\pm$ 442.1 (38) |                                                                            |
| Production rate, $r$                                                                    | 0.22 $\pm$ 0.12 (85)               | 0.07 $\pm$ 0.03 (38)    |                                                                            |
| Biomass at MSY ( $B_{msy}$ , kg/ha)                                                     | 506 $\pm$ 43.1                     | 803.0 $\pm$ 221.1       |                                                                            |
| Fished/Unfished biomass                                                                 | 0.13                               | 0.20                    |                                                                            |
| Surplus production yield estimate, MSY (tons/km <sup>2</sup> /year)                     | 5.6 $\pm$ 0.91                     | 2.8 $\pm$ 0.3           |                                                                            |
| (c) Surplus yield estimates (tons/km <sup>2</sup> /year)                                |                                    |                         |                                                                            |
| National/private park closures                                                          | 0.8 $\pm$ 1.8                      | 2.5 $\pm$ 0.9           | Same as (a)                                                                |
| Marine reserves                                                                         | 2.5 $\pm$ 0.6                      | 2.0 $\pm$ 0.5           |                                                                            |
| Community closures                                                                      | 4.3 $\pm$ 1.2                      | 2.1 $\pm$ 0.4           |                                                                            |
| Fished reefs                                                                            | 2.5 $\pm$ 0.5                      | 1.2 $\pm$ 0.5           |                                                                            |
| (d) Daily yield and revenue estimates from surplus production (kg/km <sup>2</sup> /day) |                                    |                         |                                                                            |
| Censused biomass yield estimates (kg/km <sup>2</sup> /day)                              | 11.5 $\pm$ 2.5                     | 5.7 $\pm$ 2.4           |                                                                            |
| Revenue at censused stocks (US\$/km <sup>2</sup> /day)                                  | 21.9 $\pm$ 4.7                     | 9.1 $\pm$ 3.8           |                                                                            |
| $B_{msy}$ surplus yield estimates (kg/km <sup>2</sup> /day)                             | 25.5 $\pm$ 4.1                     | 13.3 $\pm$ 1.4          |                                                                            |
| Revenue at $B_{msy}$ (US\$/km <sup>2</sup> /day)                                        | 48.5 $\pm$ 7.9                     | 21.3 $\pm$ 2.2          |                                                                            |
| (e) Fishing effort achieving living wage thresholds                                     |                                    |                         |                                                                            |
| For censused fish biomass as fishers/km <sup>2</sup> /day                               |                                    |                         |                                                                            |
| Family living wage (six persons)                                                        | 1.3 $\pm$ 0.2                      |                         | Modeled in Fig. 3 for censused and $B_{msy}$                               |
| Individual living wage                                                                  | 2.1 $\pm$ 0.3                      |                         |                                                                            |
| Poverty line                                                                            | 7.4 $\pm$ 1.7                      | 3.2 $\pm$ 0.3           |                                                                            |
| Recommended fishing effort at $B_{msy}$                                                 |                                    |                         |                                                                            |
| Family living wage (six persons)                                                        | 2.8 $\pm$ 1.7                      | 1.2 $\pm$ 0.1           | Best linear fit to age of closure data summarized in Supplementary Table 3 |
| Individual living wage                                                                  | 4.6 $\pm$ 2.8                      | 2.0 $\pm$ 0.2           |                                                                            |
| Poverty line                                                                            | 16.7 $\pm$ 9.2                     | 7.4 $\pm$ 0.8           |                                                                            |
| Annual linear production                                                                | 1.7 $\pm$ 0.50                     | 2.6 $\pm$ 0.6           |                                                                            |

Summaries are based on fishable biomass of taxa and individuals > 10 cm in length. Income thresholds are taken from government reports <https://new.knbs.or.ke/wp-content/uploads/2023/09/The-Kenya-Poverty-Report-2021.pdf> and <https://www.knbs.or.ke/wp-content/uploads/2022/05/2022-Economic-Survey1.pdf>

management systems (Fig. 1). Closure management systems in fringing reefs included 90 replicates in the three national parks of Malindi, Watamu, Mombasa, and a fourth community closure in Kuruwitu that were established as no-fishing areas in 1968 and 2006, respectively (Supplementary Table 1). Sampling in island reefs included 38 replicates in permanent national fisheries closure Kisite Marine National Park established in 1978, a private closure in Zanzibar (Chumbe Island), a county-managed closure in Pemba (Misali Island), and a well-enforced community managed closure (Wasini), which were closed

to fishing between 1994 and 1998. Repeat-sampling was undertaken and the time since closure was determined to produce the biomass recovery with the age of closure relationships. Fish censuses were also undertaken in nationally managed fisheries reserves, community closures, and fished reefs without specific restrictions apart from the national fisheries laws, which were often not enforced. The numbers of census varied from a low of 23 sites in fringing reef community closures to 122 fished reefs. Island reef samples include 47 replicates in marine reserves and 10 fished reefs (Table 1).



Fish censuses were undertaken by T. McClanahan while snorkeling and SCUBA diving at depths between 1 and 16 m and mixed reef habitats of reef slopes, crests, and back reefs. Sites were all located on calcium carbonate bottoms colonized by hard and soft corals and various algae, with sand and seagrass being a smaller portion of the bottom cover. Fish were counted in 500 m<sup>2</sup> belt transects where individual encountered fish were identified to a family and size visually estimated by standard lengths in 10 cm bins. Community biomass weights used the midpoint of the bins and family level length–weight regressions (McClanahan & Kaunda-Arara 1996). No small fish < 3 cm were counted. Larger and more mobile fish were counted first, followed by mid-sized taxa, and finally small site-attached species as to account for their potential influence by the observer.

The fishable biomass categories examined were composed of all individuals > 10 cm in 25 families with sharks and damselfish excluded due to their episodic or high variation (see McClanahan et al. 2019, 2035 for lists of taxa). The (sub)families sampled and considered for evaluations included the Acanthuridae, Aulostomidae, Balistidae, Caesionidae, Carangidae, Chaetodontidae, Diodontidae, Fistularidae, Haemulidae, Holocentridae, Labrinae, Lethrinidae, Lutjanidae (includes Nemiptheritidae), Mullidae, Muraenidae, Pempheridae, Pinguipedidae, Pomacanthidae, Rhinodontidae, Scarinae, Scorpaenidae, Serranidae, Siganidae, Sphyrnidae, and Synodontidae. Fishes < 10 cm are seldom captured in the fisheries and provide a useful cutoff for distinguishing total from fishable biomass. Removing fish < 10 cm reduced the average biomass by ~ 10 kg/ha (2.4% of total) with the largest effect in reducing Pomacentridae and Labrinae biomass. Here, we focus our presentation on fish families that were counted in census and regularly reported in the catch. Visual census methods have been vetted by many studies and found to make good estimates when larger and diver-adverse fish are counted first while laying out transect lines (McClanahan et al. 2007a, b; Bernard et al. 2013; Emslie et al. 2018).

## Fish catches

### *Landing site selection*

Data from 27 fisheries landing sites were evaluated for similarities/dissimilarities by fishing effort (fishers/km<sup>2</sup>/day), catch per fisher per day (CPUE—kg/fisher/day), catch per unit area per day (CPUA—kg/km<sup>2</sup>/day) and revenue (US\$/fisher/day) using Ward's minimum variance hierarchical clustering which employs Euclidean distance to identify unique patterns that minimizes the total within-cluster variance to produce compact and homogeneous clusters (Supplementary Table 1b). The clustering used data on effort, gear, and catch and the method minimizes the total within-cluster

variance and sets no limits to the number of cluster groups. Landing sites broadly clustered into two large categories that represented the fringing and island reef landing sites (Fig. 1; Supplementary Fig. 1). The first cluster comprised six landing sites, colored in blue, in the Shimoni-Vanga seascape characterized by island reefs. One of the sites was excluded as it lacked the full catch time series resulting in five sites for further analysis, namely Jimbo, Vanga, Kibuyuni, Wasini, and Mkwiro. The second large fringing reef cluster, colored in red and green, included 21 sites along Kenya's fringing reefs from Kilifi in the north to Gazi town in the south. Eight sites, in green, formed a distinct cluster and were most dissimilar from the island reefs and were therefore excluded. Further, seven of the 13 landing sites, in red, lacked a fully sampled time series and therefore excluded. The remaining six sites included the closely related sites in terms of proximity, geography, and fisheries attributes, namely Gazi, Mgwani, Mvuleni, Mwaepe, Mwanyaza and Tradewinds. These sites were close to the island reef sites but differing primarily in the geological fringing and island reef classifications. The final 11 landing sites were on opposite sides of a coastal elbow and therefore shared some large-scale environmental variable influences (i.e., light and oceanic water sources), while others were not shared (i.e., water flow and retention). These landing sites measured most of the catch (~90%) in the locations and were sampled independently by different observers using the same methods.

### *Fish catch sampling*

The above 11 landing sites were visited  $7.9 \pm 0.3$  (SEM) days per month between January 2021 and December 2023. Observers recorded the numbers of fishers, boats, fish groups, and weight of the catch by groups to the nearest 0.1 kg. Fish groups were derived from natural groupings that fishers used based on the market prices per group. The fish groupings included goatfish, rabbitfish, scavengers, parrotfish, and mixed catch. The mixed catch group included many species of reef-associated fish that fishers grouped together due to their shared low price. Price data per kg were collected monthly for each catch group during sampling periods and data were used to estimate revenue of fishers based on their daily catches. Landing data was entered via cellphone using Kobo Toolbox software (<https://eu.kobotoolbox.org/>), and data were downloaded into spreadsheets for analysis in R version 4.2.1. The average price of fish was US\$  $1.7 \pm 0.05$  (95%CI) although a higher price US\$  $1.9 \pm 0.04$  was realized in fringing reefs than island US\$  $1.6 \pm 0.1$ . Additionally, comparison of prices between the initial and the last three months of the study period showed significant increases ( $P < 0.003$ ). Specifically, prices not adjusted for inflation in fringing reefs increased from US\$  $1.7 \pm 0.03$ – $2.0 \pm 0.1$  and  $1.5 \pm 0.2$ – $1.8 \pm 0.1$  in island reefs.

The Kenyan shilling to US\$ exchange rate was 130 Ksh/US\$ at the time of study and we did not adjust prices for changing inflation or exchange rates. Based on interviews and landing site reports of monthly fishing effort, the number of fishing days per year was 220 and 210 in fringing and island reefs, respectively.

### *Mapping of fishing grounds*

Estimates of fishing area were undertaken using a collaborative community mapping process previously described in detail elsewhere (McClanahan and Kosgei 2023). The process is based on interviews with stakeholders using different gears (often about 15 gear types) who formed working groups. These groups created hand drawn maps that were harmonized among groups and then digitized and used in ground-truthing exercise. GPS coordinates of specific fishing grounds taken from the field visits form polygons used to finalize and estimate fishing areas. The final map is presented to community members as a final validation.

## **Data analysis**

### *Fish census analyses*

Census data were analyzed for estimating biomass in the different management areas, recovery of fishable biomass in fisheries closures, differential biomass between fished versus unfished management and fringing versus island reef sites. Mann–Whitney and Kruskal–Wallis’s statistic was used to test for differences in fishable biomass between the fringing and island reef for the four management systems.

### *Fish catch analyses*

Raw daily catch data were pooled to the month for tests of normality required for selecting statistical tests. Pooled raw and transformed monthly data failed Shapiro–Wilk (W) normality test ( $< 0.01$ ), and therefore, the Kruskal–Wallis tests were used for comparisons between pooled fringing and island sites using “kruskal.test ()” in R package “stats”. Changes over the sampling period were tested for significance by linear regression analyses for the mean values of (a) effort calculated as numbers of fishers who fished per day per area (fishers/km<sup>2</sup>/day), (b) catch per fisher per fisher per day (CPUE), as total catch divided by numbers of fishers who fished in a day (kg/fisher/day), (c) catch per km<sup>2</sup> or unit area fished in a day (CPUA as kg/km<sup>2</sup>/day), and (d) daily revenue as daily CPUE times the price per catch category (US\$/fisher/day).

### *Comparing catch sustainability models*

Several models were tested as measures of sustainability. These include standard fisheries surplus production model, an effort-catch/revenue model, fish biomass proportionality model, and a between-site rate of change model. Surplus production is based on the recovery of fish biomass after cessation of fishing and measures the natural productivity of these reefs. The effort-catch/revenue model evaluates the effects of fisher effort and incomes to estimate an economic or sustainable effort-income equilibrium. Catch proportionality is a measure of the lack of selectivity of fisheries on the fish community, where high proportionality between the natural community and catch is expected to optimize production. Finally, the between-site rate of change relationships is a method to estimate the effort and catch for which there is no change based on the zero intercept of the between-site correlations. It is a measure of no-change sustainability rather than an optimum production sustainability that is estimated from the surplus production model. Comparison of models is expected to reveal several possible sustainability metrics of each reef environment and the consequences of selecting each model.

### *Surplus production*

Total fishable biomass at the specific time since closure were used to create a recovery curve up to 54 years for the fringing and 45 years for island reefs. The three models tested for recovery rates included the asymptote, Ricker, and logistic equations. The logistic was the best fit with lower AIC, and therefore, the recovery or  $r$  and final biomass or  $B_0$  were used to estimate ecological or fishable biomass production as described below. Potential yields for the four management systems were also estimated from this procedure.  $B_t$  is the biomass in closure year  $t$ ,  $B_i$  is the initial biomass at the closure age of zero,  $B_0$  is the unfished biomass, and  $r$  is the recovery rate:

$$B_t = \frac{B_0}{1 + \left( \frac{B_0 - B_i}{B_i} \right) e^{-rt}} \quad (1)$$

Surplus MSY ecological production was then estimated for island and fringing reefs and each management type using the solution of the Verhulst equation. (Bacaër 2011).

$$\text{MSY} = \frac{rB_0}{4} \quad (2)$$

We also tested for a simpler linear recovery or annual production relationships for the by-family and total biomass in fringing and island reefs. The slope of the fit to a linear

regression is an estimate of the annual increment in biomass and net production if biomass growth is linear rather than exponential around moderate biomass values (i.e.,  $0.5 B_0$ ). This annual production method is a null or validation model to determine the efficacy of using the logistic model for estimating rates of production relative to a simpler linear or constant between-year annual production model. Comparing models can be seen as a test of the strength of the exponential portion of the logistic model. Therefore, annual production taken from the slope of the best linear fit to the closure age versus fishable biomass relationship was compared to surplus production and the estimated annual catch measured at the sampled landing sites. The R package “nlslm ()” was used to fit the models.

#### *The effort-catch/revenue model*

How many fishers could be supported in these two seascape systems for the three income thresholds for the censused biomass in fished sites and  $B_{msy}$  in fringing and island seascape reefs? To answer this question, the relationship between fishing effort and daily expected catch and revenue was modeled for the censused and  $B_{msy}$  production model for the fringing and island reefs. That is, the maximum expected daily catch per  $\text{km}^2$  divided by the number of fishers per  $\text{km}^2$  for both production at the censused biomass and  $B_{msy}$  production. Catch and revenues were aligned to determine their relationships to the Kenyan national income thresholds for fishers at the poverty, individual, and family living wages available from government publications (See Table 1).

#### *Fish biomass proportionality*

Differences in proportionality of biomass in management systems and reef systems were tested. Deviations from equal relative biomass is as an indicator of fishing selectivity and the resistance of specific taxa to fishing. Thereby reflecting deviation from the proposed “balanced harvest” management recommendations expected to maximize fisheries production (Jacobsen et al. 2014). Further, the  $r$  and  $B_0$  values derived from the above logistic model are expected to deviate from mean values as the biomass proportionality deviates from the unfished state where these values were estimated (McClanahan et al. 2025). Therefore, to test this possibility, comparisons were made of the biomass of each family relative to the total biomass in unfished closures and fished reefs. Subsequently, the relative biomass of 13 dominant and common families is presented as the percent difference from the fished state  $[(\text{fished} - \text{closures}) / \text{fished} \times 100]$  for the fringing and island reefs separately and pooled by management.

#### *Between-site rate of change relationships*

The rates of change in catch metrics within and between landing sites were used as a measure of “no-change” sustainability over the 34 months of sampling. Therefore, for each site, a linear regression of catch variables with sampling date were used to estimate each site’s mean intercept ( $b_1$ ) and slope ( $m_1$ ). For all sites in the two reef systems combined, the slopes of the best-fit model were plotted, and a best-fit linear regression determined the between-site slope ( $m_2$ ). The best-fit linear equation to the  $m_2$  slope and intercept ( $b_2$ ) was solved for a no change or zero intercept. The 95% CI of this intercept was used as a bounded measure of temporally stable sustainability for all reef type sites during the sampling period. Therefore, sustainable catch at stock biomass and environmental conditions during the sampling period could be estimated separately for effort, CPUE, CPUA, and revenue for the fringing and island reefs. This time and stock-specific stable sustainability metric differs from the ecological MSY calculated from  $B_{msy}$ . Surplus production sustainability is the predicted maximum yield at  $0.5B_0$  if  $r$  and  $K$  are constant across the fishing effort sequence.

## **Results**

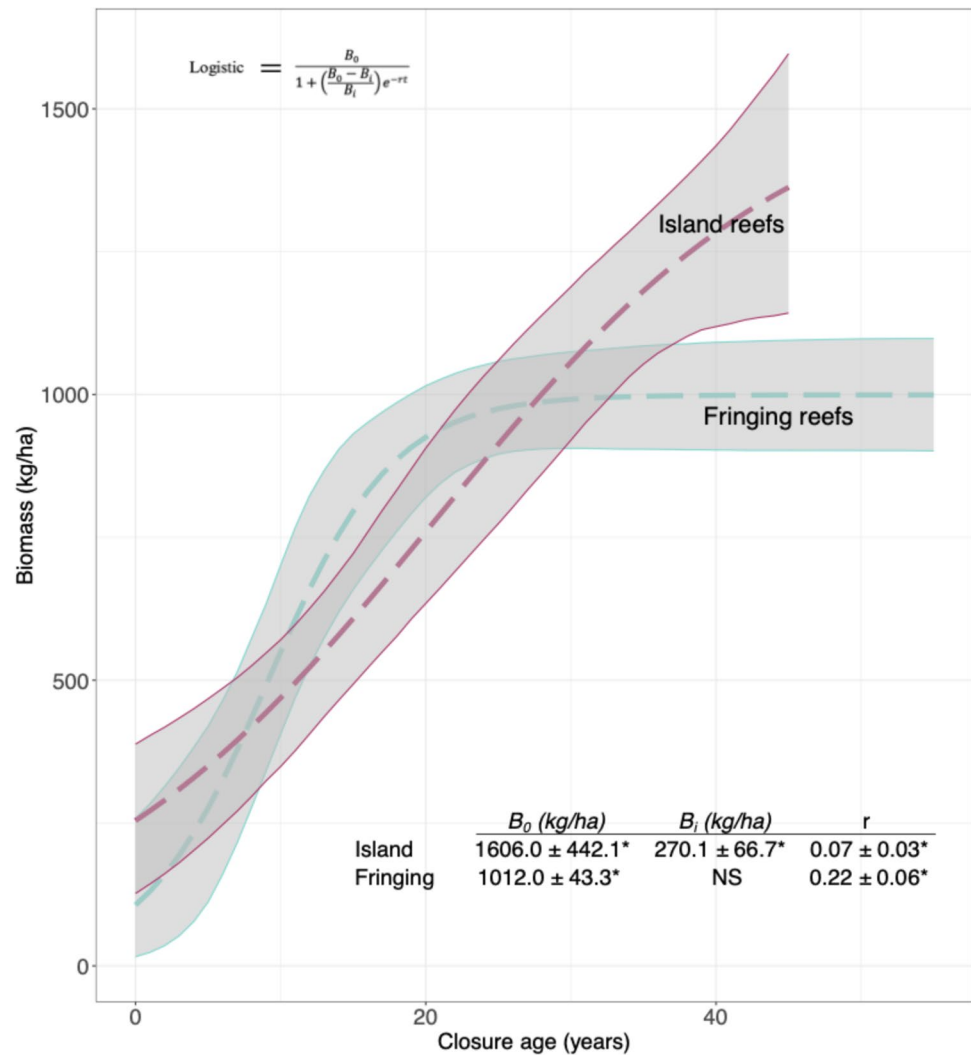
### **Fish census and recovery estimates**

Fish biomass differed among the fisheries management categories and the level of the management restrictions (Table 1a). For example, biomass from censuses in the national parks was estimated at  $\sim 1000 \text{ kg/ha}$  with no statistical differences between island and fringing reefs. Community reserves had the next highest biomass level but was lower at  $\sim 330 \text{ kg/ha}$  with no statistical differences between island and fringing reefs. Fringing reefs had lower biomasses in fished reefs and national marine reserves ( $\sim 130 \pm 26 \text{ kg/ha}$ ;  $\pm 95\% \text{ CI}$ ) than island reefs ( $\sim 385 \pm 75 \text{ kg/ha}$ ). Therefore, the fished to unfished ( $B_0$ ) biomass ratio was 0.13 for fringing and 0.20 for island reefs.

Recovery of fishable biomass estimated from the space-for-time substitution and logistic model methods suggests the equilibrium biomass of fringing was lower (1012, 95% CI = 925–1098 kg/ha) but less variable than island reefs (1606, 95% CI = 708–2504) (Table 1b, Fig. 2, Supplementary Table 2). The fishable biomass estimated production rates or  $r$  was, however, higher and more variable in the fringing (0.22, 95% CI = 0.10–0.34) than island reefs (0.07, 95% CI = 0.02–0.13). Logistic model fits at the family level indicated more variability than for the total biomass (Supplementary Table 2). Most of the significant fits were found for the high biomass families of Acanthuridae, Scarinae, Labrinae and Others while fewer low-biomass families had



**Fig. 2** Recovery of fishable biomass in East African fisheries closures. Shown are the best-fit logistic equation parameters and model (mean  $\pm$  95%CI) for a space-for-time substitution of fishable biomass in fringing (green) and island reefs (red). Best-fit equations and fit information at the family level presented in supplementary Table 2



significant fits to the logistic model depending on the variables of  $r$ ,  $B_t$ , or  $B_0$ . Low-biomass families namely Serranidae, Balistidae, Chaetodontidae and Pomacanthidae, had significant fits to the logistic model for both fringing and island reefs.

### Modeling production, effort, and revenue

Ecological surplus production MSY from the logistic surplus production model was estimated to be higher in fringing ( $5.6 \pm 0.91$  tons/km<sup>2</sup>/y) than island reefs ( $2.8 \pm 0.3$  tons/km<sup>2</sup>/y) (Table 1b). Both estimates indicated high variation possibly associated with different fish species life histories and inter-reef and habitat variation. Estimating production from biomass in the four management systems indicate the highest surplus production in community closures (2.1–4.3 tons/km<sup>2</sup>/y). Production estimates were not different between national marine reserves and fished reefs (1.2–2.5 tons/km<sup>2</sup>/y) (Table 1c). The lowest surplus production was

estimated for the old and large national closures in fringing reefs (Malindi, Watamu, and Mombasa National Parks), which indicates biomass values near the  $B_0$  equilibrium. Surplus production was higher and more variable in the more variable age and size of closures sampled among the island reefs.

Daily MSY using the  $0.5B_0$  biomass surplus production for the fringing was  $25.5 \pm 4.1$  kg/km<sup>2</sup>/day and revenue was  $48.5 \pm 7.9$  US\$/km<sup>2</sup>/day compared to the island reefs of  $13.3 \pm 1.4$  kg/km<sup>2</sup>/day and  $21.3 \pm 2.2$  US\$/km<sup>2</sup>/day. Using surplus production values, the distribution of the catch as CPUE and revenue as a function of fishing effort can be modeled for the censused and  $B_{msy}$  (Table 1d). Fisheries yield estimates for censused biomass and surplus production models reported as kg/km<sup>2</sup> per fishing day, predicted lower yields for fringing at  $11.5 \pm 2.5$  kg/km<sup>2</sup>/day than island reefs at  $5.7 \pm 2.4$  kg/km<sup>2</sup>/day. Despite the higher estimated production rate potential in fringing reefs, the low biomass resulted in estimated censused biomass production less than

the maximum potential of island reefs. Therefore, if surplus production estimates are reliable, fringing reefs were, on average, potentially losing more than twice the potential yield and revenue by reducing biomass via high fishing effort. Island reefs were estimated to be losing ~25% of the yields but exhibiting considerable between-site or island variability.

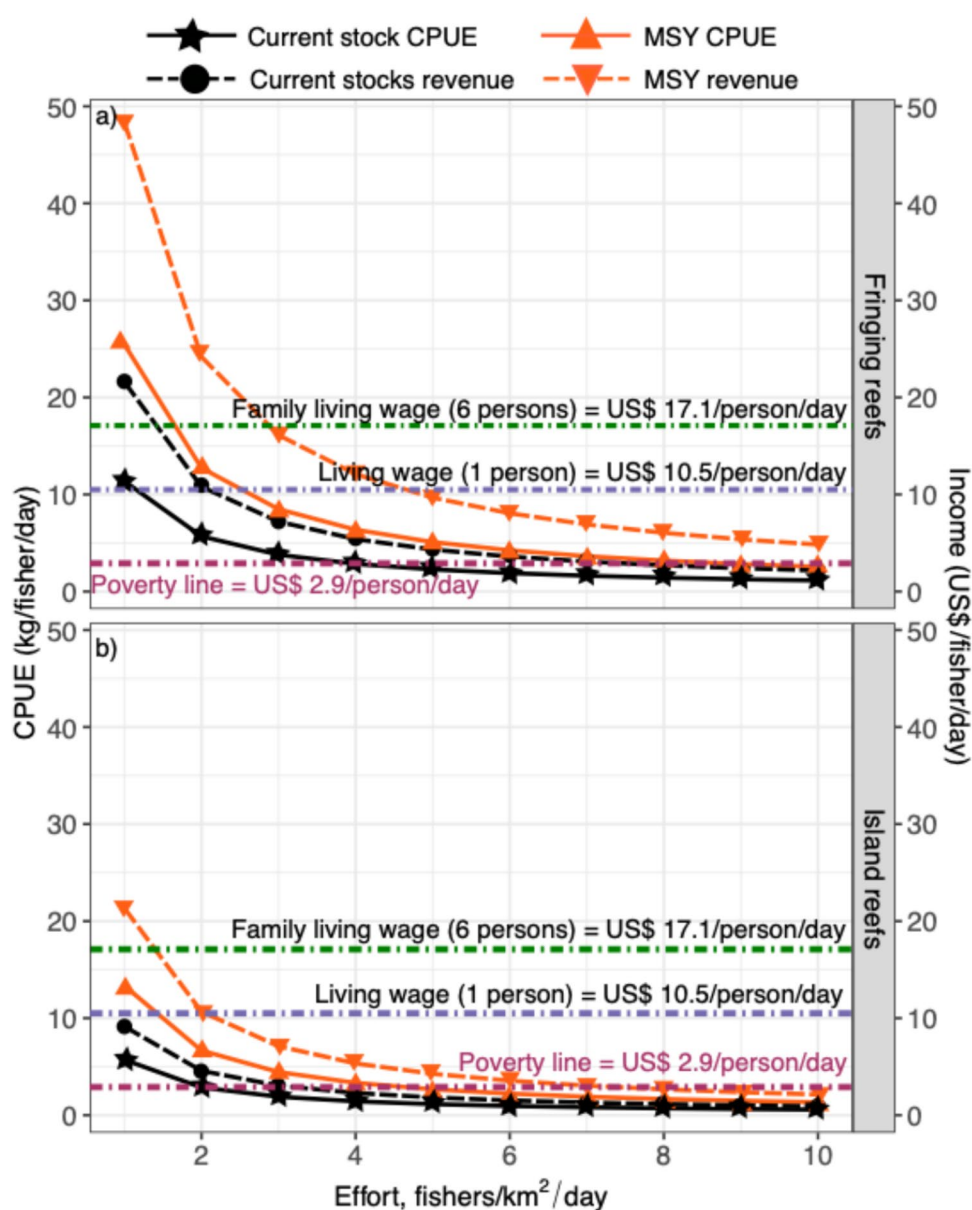
Including Kenyan national poverty, individual, and family living wage income thresholds on effort-CPUE and revenue plots provided estimates of the potential distribution of the daily production and income among fishers (Fig. 3; Table 1e). Under  $B_{msy}$  assumptions, the fishing effort thresholds for poverty, individual, and family living wages was higher for the fringing than island reefs, respectively, at 16.7 versus 7.4 for poverty, 4.6 versus 2.0 for individual, and 2.8

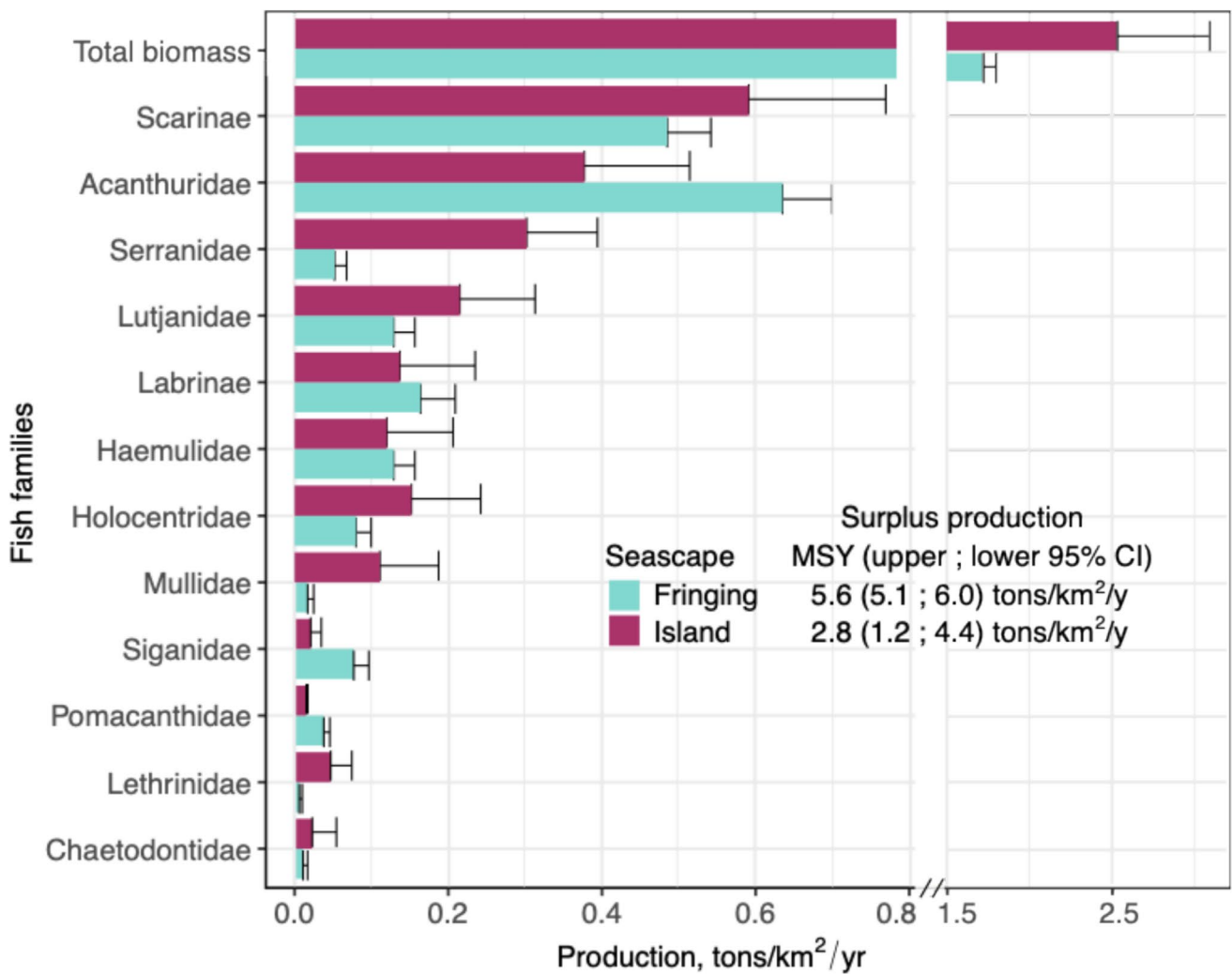
versus 1.2 fishers/km<sup>2</sup>/day for the family living wage threshold. For the censused biomass, the threshold fishing effort for fringing and island reefs was similar for the three revenue thresholds at ~9.3, 2.6, and 1.6 fishers/km<sup>2</sup>/day, respectively. Consequently, surplus ecological production models predict fringing reefs have greater potential to support more fishers and households than island reefs when  $B_{msy}$  is achieved.

### Annual production of fish taxa

For the studied families, annual production estimates indicate more significant fits to the linear than the logistic models (Supplementary Table 2 and 3). Herbivorous Scarinae and Acanthuridae are the largest contributor to production, followed by mixed carnivorous Serranidae and Lutjanidae

**Fig. 3** Modeled daily per fisher catch and revenue as a function of fishing effort. Predictions are based on the censused and  $B_{msy}$  as daily potential yields for in **a** fringing and **b** island reefs. Daily yield in the fringing reefs for surplus ecological production estimate of  $MSY = 25.5$  kg/km<sup>2</sup>/day compared to 11.5 kg/m<sup>2</sup>/day for censused biomass. Daily yields for island reefs for surplus production estimate of  $MSY = 13.3$  kg/km<sup>2</sup>/day and under measured biomass of 5.7 kg/m<sup>2</sup>/day. See Table 1. Shown are Kenyan official national thresholds for poverty, individual, and family livable wages thresholds in 2020





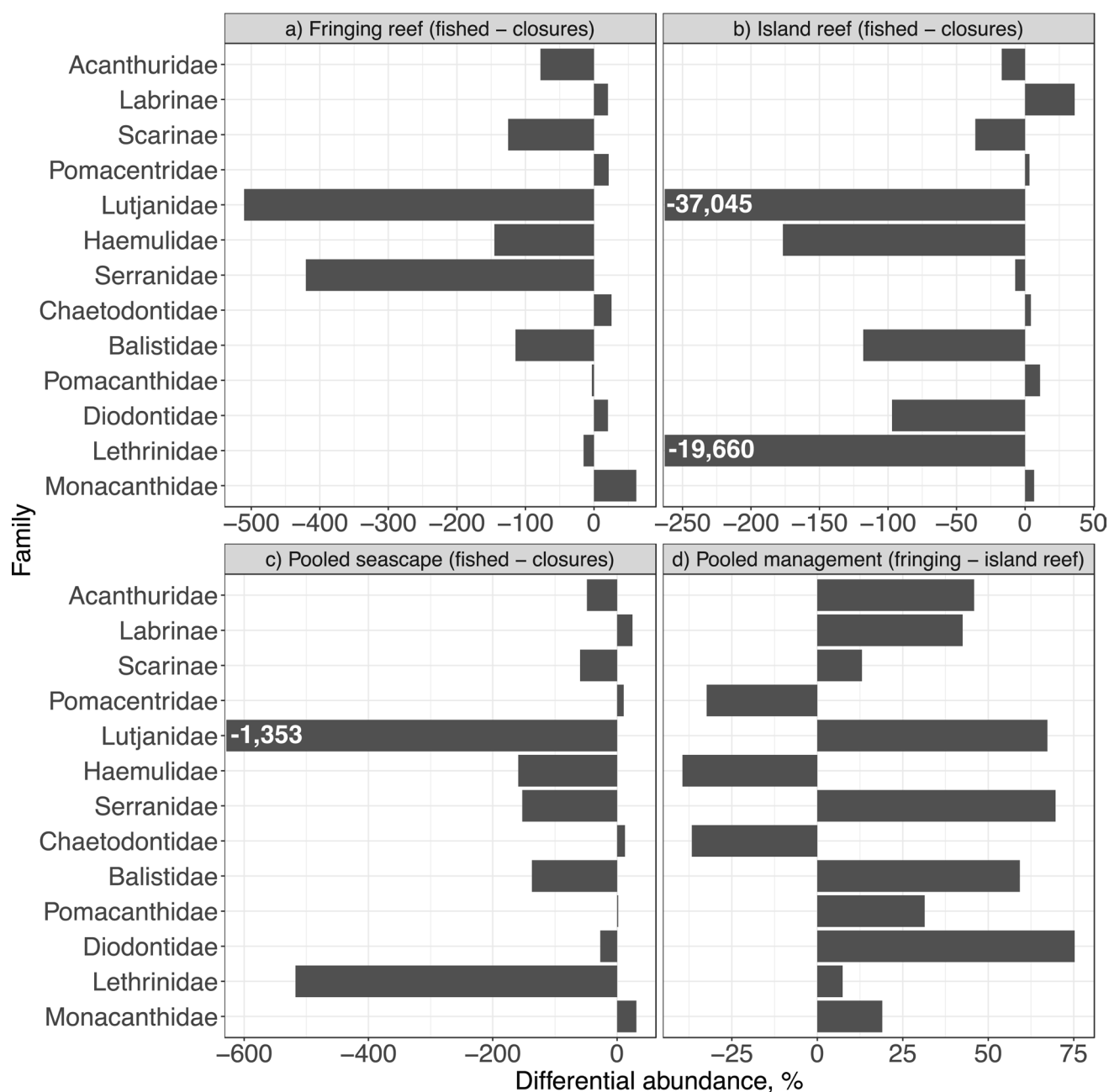
**Fig. 4** Linear annual fish production rate estimates from fisheries closures in fringing and island reefs. Total and family level annual production estimates (mean  $\pm$  95% CI) based on the annual rates of changes in biomass in fisheries closures in fringing (green) and

island reefs (red). MSY derived from surplus production estimates (Table 1b) was presented in the legend. Fish families are ordered from estimates of most to least production based on the linear fit intercepts. Full statistical details given in supplementary Table 3

and several low-biomass taxa (Fig. 4). Higher significance and stronger fits were seen at the family level in the fringing than island reefs (Supplementary Table 3). However, the Serranidae, Balistidae, Lethrinidae, Mullidae and Labrinae showed the opposite pattern or weaker fits in the fringing than island reefs. Nevertheless, the linear per year production method indicated lower annual than surplus production estimates, with differences being largest in the fringing reefs. For example, the total annual fringing reef production estimate was  $1.7 \pm 0.50$  tons/km<sup>2</sup>/y and well below the 95% CI of the logistic surplus production model predictions of  $5.6 \pm 0.91$  tons/km<sup>2</sup>/y. Island reefs' annual production estimates was  $2.6 \pm 0.6$  tons/km<sup>2</sup>/y but within the 95% CI prediction intervals of the surplus production model of  $2.8 \pm 0.3$  tons/km<sup>2</sup>/y. This comparison indicates stronger exponential growth potential in the fringing compared to island reefs.

### Taxa proportionality differences

Proportionality of taxa evaluations indicated large relative differences in the abundance of some key fished taxa between unfished and fished locations (Fig. 5). Both fringing and island reefs exhibited large losses of proportionality among the Lutjanidae, Lethrinidae, Haemulidae, Balistidae, Scarinae, and Acanthuridae attributable to fishing (Fig. 5a). The island reef analysis indicated large losses of the Haemulidae and Lethrinidae in fished reefs whereas the fringing reefs had more pronounced losses among the Serranidae (Fig. 5b). Pooling sites by reef systems indicated the overall fishing resilient taxa were Labrinae, Pomacanthidae, Chaetodontidae, Monacanthidae, and Pomacanthidae (Fig. 5c). Comparing proportionality differences between fringing and island reefs irrespective of



**Fig. 5** Differential abundance (%) of fish families censused in fisheries closures and fished reefs. The relative numbers of individuals among the fish families common to the census in fisheries closures and fished reefs in fringing and island reefs are observed. Differentials show differences in numbers of individuals in fished reefs and closures from field transects  $[(\text{fished} - \text{closures}) / \text{captured}] \times 100$  in

**a** fringing, **b** island reefs, and **c** both reefs types combined. **d** differential abundance for fringing and island reef pooling data into fished and unfished closure management  $[(\text{fringing} - \text{island}) / \text{fringing}] \times 100$ . Families ordered from most to least abundance estimates from all field census

management indicated that fringing reefs had greater relative abundance of most taxa apart from the Haemulidae, Chaetodontidae, and Pomacentridae, which were relatively more abundant in island reefs (Fig. 5d).

#### Fisheries catch estimates and time series

Analyses of fishing landing data found that fishing effort and CUPA were higher in fringing than island reefs (Table 2a,

**Table 2** Summary of fisheries landing data for (a) effort, catch, and revenue estimates, (b) annual rates of change in these estimates based on slopes of monthly average catches over a ~34-month period, and (c) the equilibrium of between-site zero-intercept values of these catch metrics (see Fig. 6)

| Fisheries catch landings information    | Fringing reef | Island reef |                |
|-----------------------------------------|---------------|-------------|----------------|
| (a) Estimates                           |               |             |                |
| Effort (fishers/km <sup>2</sup> /day)   | 4.2 ± 0.4     | 1.3 ± 0.1   | 214.2; <0.0001 |
| CPUE (kg/fisher/day)                    | 3.8 ± 0.1     | 4.1 ± 0.2   | 5.9; 0.02      |
| CPUA (kg/km <sup>2</sup> /day)          | 15.0 ± 1.6    | 5.3 ± 0.7   | 157.8; <0.0001 |
| Revenue (US\$/fisher/y)                 | 7.0 ± 0.3     | 6.6 ± 0.5   | NS             |
| Fish prices, US\$/kg                    | 1.9 ± 0.04    | 1.6 ± 0.1   | 55.2; <0.0001  |
| (b) Slope of the annual rates of change |               |             |                |
| Effort (fisher days/km <sup>2</sup> /y) | -11.1 ± 8.6*  | -1.1 ± 2.0  | 214.2; <0.0001 |
| CPUE (kg/fisher/y)                      | 0.9 ± 4.3     | 3.0 ± 4.7   | 5.9; 0.015     |
| CPUA (kg/km <sup>2</sup> /y)            | -31.0 ± 28.6* | 0.7 ± 12.0  | 157.8; <0.0001 |
| Revenue (US\$/fisher/y)                 | 11.0 ± 8.6*   | 17.9 ± 9.0* | NS             |
| (c) Equilibrium values, zero intercepts |               |             |                |
| Effort (fishers/km <sup>2</sup> /day)   | 2.5 ± 2.0     | 1.1 ± 0.3   | See Fig. 7     |
| CPUE (kg/fisher/day)                    | 3.8 ± 0.5     | 4.3 ± 1.0   |                |
| CPUA (kg/km <sup>2</sup> /day)          | 10.2 ± 5.0    | 5.3 ± 1.9   |                |
| Revenue (US\$/fisher/day)               | 8.9 ± 1.1     | 8.8 ± 5.1   |                |

\*Slopes different from zero at  $p < 0.005$  by linear regression

Supplementary Table 4). Specifically, effort being more than three times higher in in fringing reefs ( $4.2 \pm 0.4$  fisher/km<sup>2</sup>/day) than island reefs ( $1.3 \pm 0.1$  fisher/km<sup>2</sup>/day). Similarly, CPUA was higher in fringing ( $15.0 \pm 1.6$  kg/km<sup>2</sup>/day) than island reefs ( $5.3 \pm 0.7$  kg/km<sup>2</sup>/day). However, CPUE was not significantly different between reef types and both around ~4 kg/fisher/day. Revenue was significantly but marginally higher in fringing ( $7.0 \pm 0.3$  US\$ per day) than island reefs ( $6.6 \pm 0.5$  US\$ per day) associated with higher per kilogram prices in fringing ( $1.9 \pm 0.04$  US\$/kg) than island reefs ( $1.6 \pm 0.1$  US\$/kg). Prices rose similarly in both the fringing and island reefs over the three year study period at ~0.15 US\$ per year.

Time series of monthly average effort, catches, and revenue over ~34 months indicated more stability for these metrics in the island than fringing reefs (Fig. 6). In fringing reefs, effort and CPUA declined significantly, while CPUE did not change (Table 2b). In island reefs, there were no statistically significant changes in effort, CPUE, and CPUA. However, revenue rose significantly over time in the island reefs from between poverty (2.9 US\$ per day) and the individual living wage gap (10.5 US\$ per day) to around 6 to 7 US\$ per day. The rising revenues were associated with a 19% rising annual price per kilogram of fish in the island and a 15% annual rise in fringing reefs.

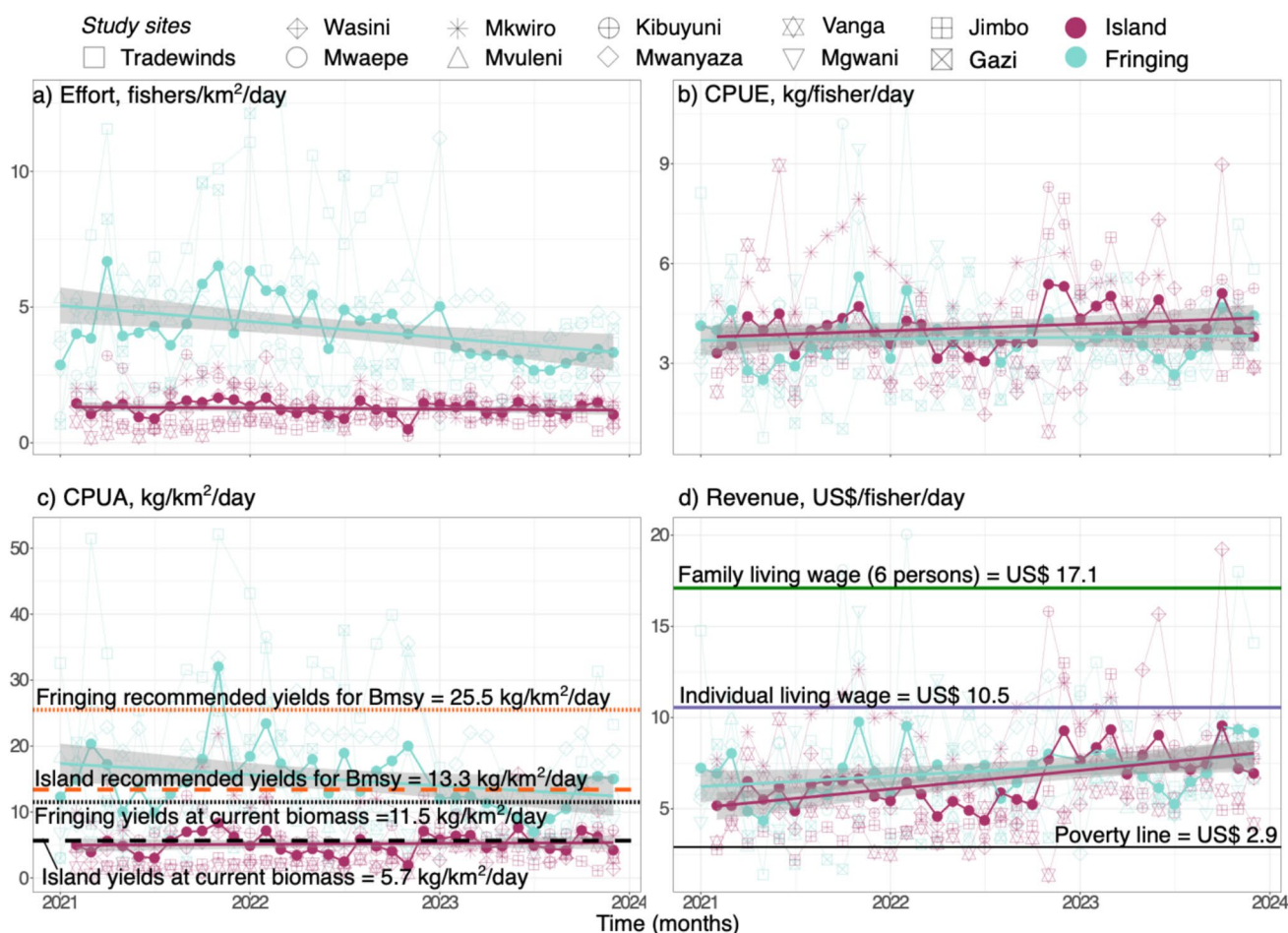
### Quantity-rate dependency and sustainability of censused biomass

Between-site quantity-rate of change relationships all show declines or negative density dependence. This is shown in the best-fit linear models of the between-site negative

relationship of the mean estimate of effort, CPUE, CPUA, and revenue when comparing landing site catch statistics (Fig. 7). Differences between fringing and island reef systems differed in that effort and CPUA in individual landing sites of island reefs had a mixture of rising and declining changes while all fringing reefs were either not changing or declining. CPUE showed more overlap between the two reef systems and more variability in the positive and negative directions of the change. Change in revenue declined with increases in mean prices and fringing and islands showed overlapping values. All individual landing site change was positive or above the no-change line, which is consistent with the rising prices of fish and revenue.

Between-site best-fit equations indicated several differences and similarities between the two reef systems for estimates of the no-change or stable sustainability of censused stocks (Table 2c). For example, the zero intercept and 95% CI windows suggest sustainable effort of censused biomass lay between  $2.5 \pm 2.0$  fishers/km<sup>2</sup>/day in fringing and  $1.1 \pm 0.3$  fishers/km<sup>2</sup>/day in island reefs. Sustainable levels of fish catch were estimated to be between  $3.8 \pm 0.5$  kg/fisher/day in fringing and  $4.3 \pm 1.0$  kg/fisher/day in islands. The expected daily per area sustainable yield was predicted to be between  $10.2 \pm 5.0$  kg/km<sup>2</sup>/day in fringing and  $5.3 \pm 1.9$  kg/km<sup>2</sup>/day in island reefs. Regardless, equilibrium revenues per fisher were more similar at  $8.9 \pm 1.1$  US\$/day in the fringing and  $8.8 \pm 5.1$  US\$/day in islands. Evaluations of per area production from  $B_{msy}$  estimates were considerably higher than censused stocks stable sustainability. The other  $B_{msy}$ -derived fisheries metrics of CPUA, CPUE, and revenue depended greatly on the fishing effort at MSY.





**Fig. 6** Time series plots of the monthly means **a** fishing effort, **b** catch per unit effort (CPUE), **c** yields as catch per unit area (CUPA), and **d** revenue for the period January 2021 to December 2023 in six fringing (green lines) and five island reef (red lines) fish landing sites in southern Kenya. Linear best-fit lines represent mean of all landing sites and the best-fit linear equations and 95% CI. **c** the estimated

CUPA for fringing and island reefs based on the surplus production for censused and estimated  $B_{msy}$  (see Table 1b). **d** horizontal lines represent Kenya's published national revenue thresholds for poverty, an individual, and a family living wage revenue. One US\$ = 130 Kenya Shillings at the end of July 2024

### Comparison of production estimates

Comparison of production by the different methods indicated how methods affect estimates and their variability (Table 3). Comparisons between methods were made possible by multiplying fishing days by the daily catch rates and putting values on the same common fisheries catch units of tons/km<sup>2</sup>/y. Fringing reef estimates indicate overlap in the surplus production and zero-intercept equilibrium methods. However, actual measured catches were above and annual linear production below zero-intercept predictions, which indicates a fisheries production deficit or overfishing relative to surplus production methods. Island reef estimates of catch production indicated overlap in estimates with the no-change or zero-intercept estimates. However, surplus and annual production overestimated the actually landed catches. This suggests a missing source of productivity likely attributable

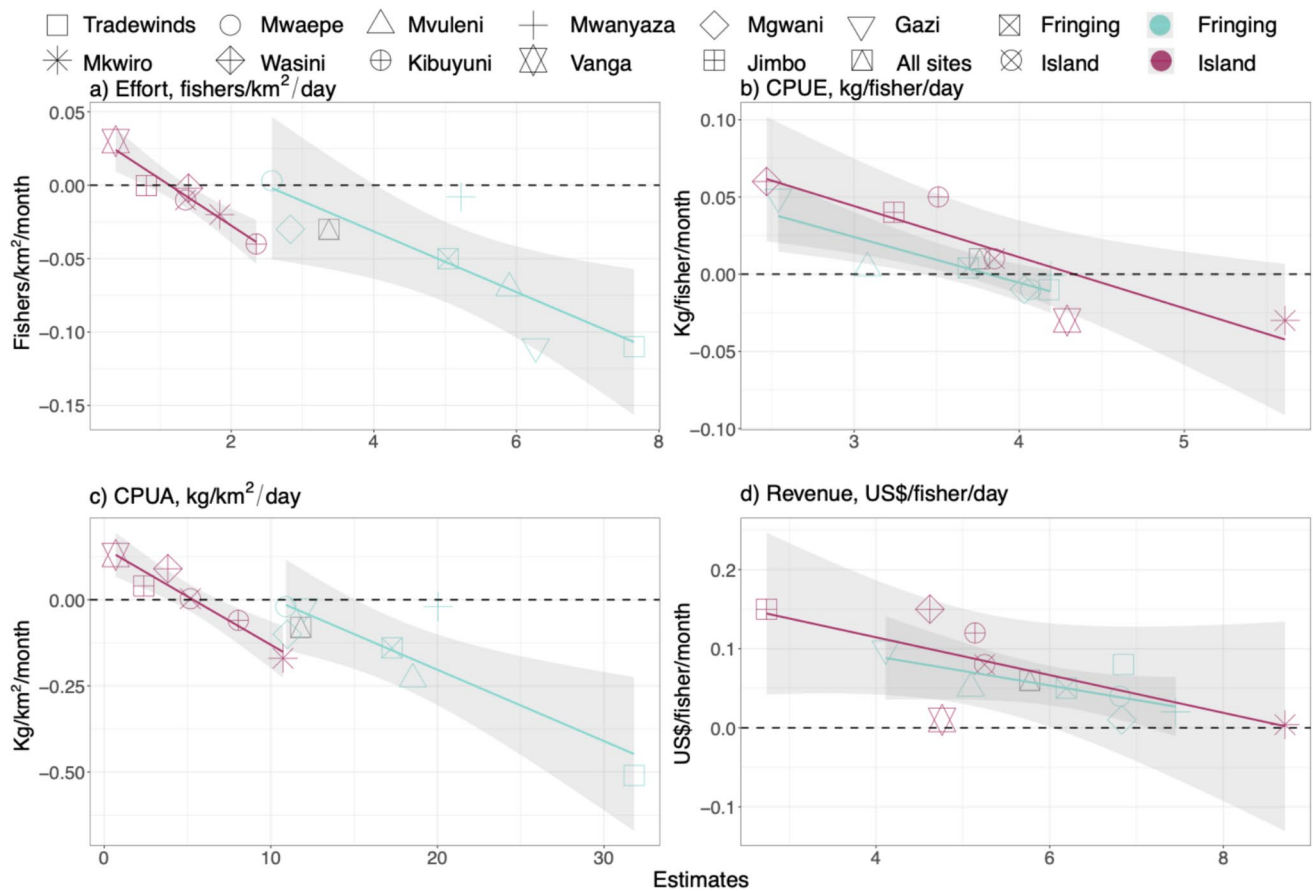
to the large disproportionate losses of taxa with high biomass but low to moderate production.

## Discussion

### Ecological and fisheries models

#### Variability in yields among reef systems

Differences in environments, reef types, productivity, potential and realized yields, rates of decline as a function of effort and yield, and income thresholds reveal complex social–ecological context and interactions that influence yields and potential estimates of sustainability. The composite findings show the importance of making many measurements and evaluating disparate methods, metrics,



**Fig. 7** Tests of density dependence. Plots of the linear changes among the studied landing sites plotting the individual landing site's estimates (intercept) and slopes for the 4 fisheries catch metrics. Estimates and slopes are based on the sites time series collected and

analyzed for a 34-month period. Plots show the best-fit linear models and 95% CIs for between-site slope-intercept relationships for **a** fishing effort, **b** CPUE **c** CPUE, and **d** revenue in fringing (green) and islands reef (red) landing sites

**Table 3** Comparison of production rates (tons/km<sup>2</sup>/year) for the methods used to estimate fisheries catch for the censused fish biomass

| Comparison of production for censused biomass                                    | Fringing reef | Island reef |
|----------------------------------------------------------------------------------|---------------|-------------|
| Fisheries landed catch production, CPUE (tons/km <sup>2</sup> /year)             | 3.30 ± 0.35   | 1.11 ± 0.15 |
| Surplus logistic production for censused biomass                                 | 2.53 ± 0.55   | 1.30 ± 0.51 |
| Annual linear production in fisheries closures                                   | 1.70 ± 0.50   | 2.60 ± 0.30 |
| Rate of change zero-intercept catch production from between-site rates of change | 2.24 ± 1.10   | 1.11 ± 0.40 |
| Ratios of production/censused biomass ratios                                     |               |             |
| Fisheries landed catch, production/biomass                                       | 0.25          | 0.12        |
| Surplus production/biomass                                                       | 0.20          | 0.06        |
| Annual production/biomass                                                        | 0.13          | 0.12        |
| Fisheries zero-intercept catch production/biomass                                | 0.17          | 0.05        |

Daily catch was converted to annual yields based on 220 and 210 fishing days on fringing and island reefs, respectively

and locations to contextualize factors affecting fisheries production before concluding about sustainability status (Pauly et al. 2013). For example, the logistic or surplus production model derived from biomass recovery in fisheries

closures allowed an estimate of biomass specific production. Moreover, fisheries catch production options indicate several possible methods to calculate production estimates and sustainability. This creates challenges for making confident

estimates without considering the assumptions and long-term consequences for fish and fisheries production.

Differences in environmental variability, taxonomic composition, and fishing effort influenced model predictions and yields. Specifically, through differential reduction in some key target taxa, notably snappers, emperors, sweetlips, and groupers. Therefore, differential resilience among taxa potentially created an imbalance in the biomass distributions and production estimates. Consequently, missing biomass-catch proportionality targets and reducing the accuracy of capture estimates based on natural whole community recovery  $r$  estimates. Beyond the findings of this study, fisheries capture models that assume static effort-yield relationship (“equilibrium models”) further overestimate production compared to nonequilibrium surplus production models (McClanahan and Azali 2020). Inaccuracies increase as the per taxa mismatch between capture and ecological fish production grows (Fig. 7). Therefore, because static models fail to parametrize the long-term taxonomic and biomass depletion rates, they can poorly estimate long-term optimal effort and catch. In other words, short-term selective overharvesting can lead to long-term changes in production potential and inaccurate predictions.

Comparing the between-site rates of change and the high variation in monthly catch rates among landing sites provided insight into the standard surplus model’s efficacy. In the island reefs, the surplus production, zero-intercept estimate, and landed catch fell within a 95% CI of  $\sim 0.7$ – $1.8$  tons/km<sup>2</sup>/year at an estimated fishable biomass of  $\sim 21$  tons/km<sup>2</sup> (Table 3). This suggests that the recovery rate estimated of  $0.07 \pm 0.03$  from closures recovery is a fair but not accurate predictor of catch. Overharvesting relative to ecological production is, however, likely to produce a small overestimate bias in landed catch. In contrast, the linear annual change method appeared to overestimate production, which is likely due to net migration of fish into closures, which will overestimate production relative to the surplus production assumption of fully endogenous production. Capture selectivity is also expected to bias surplus production methods based on the production/biomass of fishing resilient fish taxa. Increased turnover among resilient taxa can compensate for reduced biomass of slow turnover taxa in some fisheries systems (Morais et al. 2023). However, if key target taxa are reduced greatly or potentially lost from the fishery, the result is a reduction in community production and an overestimate of MSY. How these two forces emerge and affect sustainable fisheries estimates may depend on environments, social–ecological, and gear use contexts that are not well understood.

Yields differed between fringing and island reefs attributable to interactions between geographic and environmental and associated taxonomic and life history differences. Taxonomic composition was likely more influential in island

than fringing reefs. For example, even with some lost proportionality of taxa in the fringing reef there was good correspondence between the surplus and zero-intercept production methods. Fewer losses of proportionality among the snappers, emperors, and grunts in fringing than found for the island reefs likely explain this correspondence and high variability among sites. The higher loss of grouper proportionality in fringing compared to island reefs is expected to have a small affect on total yields because of their low biomass. However, the higher biomass and production taxa including surgeonfish, wrasses, and rabbitfish likely compensated for some lost production in fringing reefs. Therefore, fringing reef production rate estimates ( $r = 0.22 \pm 0.12$ ) are likely to be fair but not accurate estimates of production. Nevertheless, it is likely that spatial and temporal variability in coral reefs is too high to make highly accurate predictions of yields. Large differential proportionality findings here generally support local extinction studies where several taxa in the grunts and grouper families were found to be rare or extirpated in Kenya (Buckley et al. 2019; Otway et al. 2025). Consequently, restoring these rare taxa in the fished reefs and not just the closures is needed to recover proportionality and ecological production.

## Human dimensions of fishing

### *Declines in fishing effort*

The rates of decline in fishing effort in fringing reefs could potentially reduce the observed overfishing effects. Specifically, an annual decline of  $11.1 \pm 8.6$  fisher days/km<sup>2</sup>/y was highly variable at the studied fringing reef landing sites. The negative effort-decline relationship showed that the more the yields surpassed sustainable yields, the faster the decline in annual fishing days. CPUE was likely stabilized by this decline in effort. The recorded 0.03 tons/km<sup>2</sup>/y decline in catches is likely less than an earlier estimates made between 2001 and 2016 of 0.05 tons/km<sup>2</sup>/y (McClanahan 2018a). Fishers have increasingly reduced their numbers and daily effort since 2001 to produce this decelerating decline. Nevertheless, rising fish prices likely attenuated the exiting rate as fisher’s daily revenue rose by  $\sim 0.5$  US\$ per year. Island reefs have not been previously studied for rates in change. Nevertheless, during this more stable period, the increase in revenue among island fishers was double that of the fringing reef fishers and therefore possibly slowing fisher’s effort reductions and exiting rates. Recovery of target taxa is likely to be suppressed by slow exiting rates.

Fishers resistant to reducing effort maintained their revenues in the declining fisheries as fish prices rose. Consequently, estimates of yield using CPUE multiplied by the numbers of fishers commonly used in catch reconstructions (LeManach et al. 2015) are expected to hide yield declines

and overestimates of yields in declining fisheries. Many fishers remain in fisheries with yields below the potential maximum (Giron-Nava et al. 2021). In this Kenyan case study, fisher's persistence was maintained, as reflected by the stability of CPUE and revenue in both systems, which remained between the poverty and minimum individual national income thresholds. This was evident in both fringing and island reef landing data as the variation in effort matched the differences in ecological productivity. Fisher's resilience and retention dynamic is likely to be a pervasive dynamic in tropical countries (Teh et al. 2024). Kenyan fisheries are not the poorest of tropical nations and the poverty threshold dynamic is likely to be even more pronounced in other national reef fisheries depending on the relative cost of living and fish prices.

The temporal context of this study is provided by two previous studies. The earliest study undertaken between 1996 and 2006 reported fishing effort of  $\sim 13$  fishers/km<sup>2</sup>/day (ranging by landing site between 5 and 30 fishers/km<sup>2</sup>/day (McClanahan et al. 2008). The reported range of per area yields was between 5 and 18 tons/km<sup>2</sup>/y depending on the fishing effort and gear. A 2001 to 2016 study reported yields of  $6.6 \pm 5.1$  tons/km<sup>2</sup>/y for a fishing effort of  $\sim 15$  fishers/km<sup>2</sup> (McClanahan et al. 2018a). Yields declined over this 15 year period at a rate of 0.05 tons/km<sup>2</sup>/y for a no-change definition of sustainability of 3.9 tons/km<sup>2</sup>/year at 4.4 fisher/km<sup>2</sup> and a stock biomass estimates between 20 and 33 tons/km<sup>2</sup>. The surplus production model suggested that if biomass were increased, the estimated yield would be 6.4 tons/km<sup>2</sup>/year at 10.9 fishers/km<sup>2</sup>. Therefore, viewed over this longer study period, no-change sustainability had further declined to  $2.2 \pm 1.1$  tons/km<sup>2</sup>/year at an effort of  $2.5 \pm 2.0$  fishers/km<sup>2</sup>/day for a biomass of 13.0 tons/km<sup>2</sup>. The effort, CPUE, and no-change MSY estimate have therefore been declining for over 30 years, which is reported more broadly for many fisheries (Edgar et al. 2024). A lack of strong negative feedback between effort and fisher's incomes driven by rising prices, appears to have promoted the decline in CPUE.

### *Reef fisheries economics*

Our findings align well with a price feedback model that proposes that rising fish prices maintain overfishing and lead to a stock collapses (Dao et al. 2023). Lowered costs of boats, gear, and other extractive subsidy policies likely promote effort retention and the fisheries collapse in Kenya (McClanahan and Kosgei 2019). This is often exacerbated by welfare-based extraction subsidies that slow the exiting rate from underperforming fisheries (Daw et al. 2012; Skerritt et al. 2023). Fishing effort has been declining over many years, but has not declined rapidly enough to allow the recovery of biomass to  $\sim 50$  tons/km<sup>2</sup>/y needed to achieve the ecological  $B_{\text{msy}}$ . Recovery of biomass is expected to take

a number of years of additional reduced fishing effort to restore biomass and taxa that no longer common in the fishery (McClanahan et al. 2016).

The two reef systems showed evidence for a per person catch market equilibrium that lay between the national poverty and minimum individual income thresholds. This dynamic occurred despite differences in potential productivity of the two reef systems. Consequently, fishing effort would appear to be self-regulated around a minimum acceptable or individual fishers' opportunity cost or living wage decisions. If regulated for maximum per area yield or revenue, then we would observe effort to align more with biomass and productivity than income. An economic or opportunity cost equilibrium is expected when unrestricted access, weak regulation, and welfare-based policies are combined (Giron-Nava et al. 2021; Dao et al. 2023). Restoring fish biomass to the ecological MSY is likely to be opposed by the counter forces of rising fish prices and subsidies.

The number of days that fishing needs to be reduced to eliminate the biomass deficit for the measured biomass can be estimated from our findings (Table 3). For example, the number of fisher days (fisher/km<sup>2</sup>/day x fishing days per year) required to balance production and capture for the fringing reefs is  $374 \pm 88$  fisher days or 40% of the measured rate. Island reefs would need to reduce effort by  $42 \pm 22$  fisher days or 15% of the measured rate to be sustainable. Fishing effort could be reduced to these numbers by higher compliance with existing fisheries management regulations.

The global pervasiveness of biomass reduction suggests that compliance with fisheries regulations to increase individual and community wealth are less common than decisions to provide welfare (Bene et al. 2010; Teh et al. 2024). In welfare-based fisheries, the flexible and subsidized behaviors of fishers allow them to provide some basic household income. We see here, however, measured income is less than needed to support an individual or family without supplemental incomes. However, even women fish traders who contribute to male fishers' incomes to support their households, have insufficient incomes to meet their basic needs (McClanahan and Abunge 2017). Therefore, both male and females in these fisheries' dependent households face similar marginal incomes and rising cost-of-living problems.

The high employment and welfare policies that dominate tropical fisheries ensure that nearshore fisheries are a source of employment and food security for many multi-livelihood, economically marginalized, and informally employed people (Bene et al. 2010; Basurto et al. 2025). A perceived benefit of the flexible welfare approach is that noncommercial and often open-access fisheries provide employment and resources to 53 million or 45% of the fisheries workforce globally (Virdin et al. 2023, data from 2016). However, our findings show that this open-access and the market supply and demand approach eventually erodes the long-term



capacity to absorb the un- or underemployed and provide food, while reducing biodiversity and associated food production (McClanahan 2022). Efforts to quantify the social welfare benefits of open-access fisheries seldom consider long-term changes and the rate that socio-ecological metrics degrade. Additionally, rising prices are expected to reduce the amount of home consumption as wealthier consumers are willing to pay the higher prices. Fishing households often use this income to purchase cheaper and less nutritious foods.

### Recommendations

Our effort-CPUE-living wage model suggests that unrestricted access behavior will lead to similarly low employment capacities even where ecosystems have different production potentials (Fig. 3). Specifically, we found similar and low employment capacities when comparing the more productive fringing and less productive island fisheries. Differences arise in fishable biomass and P/B ratios to produce different effort–yield–revenue equilibriums. If ecological MSY is achieved, the higher productive systems can create more employment and food but only by reducing fishing effort or increasing fish biomass. Regardless of the differences in production, both reef systems maintained fishers at poverty levels at  $\sim 7.5$  fishers/km<sup>2</sup>/day for the measured stocks, but this is predicted to rise to 16.7 fishers/km<sup>2</sup>/day in the fringing reef at  $B_{\text{msy}}$ . Thus, achieving a surplus production MSY capture goal can create more long-term employment and support more fishers while supporting biodiversity goals.

In our case study, the fringing reef's daily effort needs to be reduced by 40% while the island reef effort needs to be reduced by 15% to be sustainable. As stocks and production increase, this temporary loss of employment can eventually be recouped. Moving closer to the benefits of MSY production and employment goals should be possible by finding acceptable means to adjust effort, compliance, and promote alternative sources of employment (Cunnigham et al. 2009). Stock recovery programs have been shown to be one of the most successful management tools if the required governance can be achieved (Melynchuk et al. 2021). Therefore, the appropriate balance between resource protection and extraction subsidies and pricing that balances entry and exiting rates with production potential should be considered among the effective tools to promote sustainability.

Protecting biodiversity and natural ecosystems is expected to have several benefits for sustainability. Therefore, we recommend that surplus models should guide management while recognizing the environmental and fish life history effects on variability in production (McClanahan 2022). However, we also expect the values and goals

of the fisheries stakeholders will influence the goals and methods chosen to estimate sustainability. Most welfare-based policy decisions will weigh the benefits of employment, revenue, and trade maximization with production potential and ecological integrity. However, these methods seldom consider the appropriate living wage thresholds and the consequences for fisher households. We suggest social combined with ecological sustainability thresholds be used as indicators of social–ecological sustainability. In the absence of high compliance with national regulations, the poverty-individual living wage window has become the default self-regulatory negative feedback mechanisms in both fringing and island systems. This revenue window is likely to marginalize fisher households while leading to overfishing and lost employment potential.

Biomass was reduced more in fringing than island reefs despite the opposite patterns in ecological production. Consequently, individual decisions and weak compliance with national fisheries laws have yet to regulate yields and revenues near their maximum production potential. This is likely to be a common outcome in many small-scale fisheries, which cannot be rectified solely with local resource protection subsidies (Teh et al. 2024). Numbers of fishers and demand for fish are simply greater than fish production on large scales and leading to considerable losses of employment, food, and revenue in poor tropical countries (Zeller et al. 2020; McClanahan et al. 2023). Therefore, the individual fishers' minimum income decisions will need active management and compliance if community and national resources are to be increased to maintain higher yields and biodiversity. We recommend promoting a fisheries goal of achieving livable wages as a key part of fisheries management deliberations. When living wage are achieved, so are more biodiversity and production potential goals (McClanahan 2022). Finally, reef ecosystems are expected to be more resilient to climate change and produce more food and employment when managed for moderate levels of fish biomass (McClanahan and Muthiga 2016; Gove et al. 2023; Ford et al. 2025).

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**Author contribution** T.M. conceptualized the study, raised research funds, checked the data, and wrote the manuscript. J.K. collected the data, analyzed the data, prepared figures, and edited the paper.

**Data available** <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/KKCHUV&version=DRAFT>.



## Declarations

**Conflict of interest** The authors declare no competing interests.

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